Ecotox & Marley Team Charter

Alexa, Peyton, Marley

1. Team Purpose or Mission

What does your team hope to accomplish by the end of the project? What questions will your dashboard help users explore, and who is the intended audience? How will your combined skills and interests support the work? What can your audience expect in terms of analytical quality, clarity, and professionalism?

Our team aims to evaluate freshwater plant mortality as a result of heavy metal exposure from the combustion of fossil fuels. Lead, arsenic, copper, cadmium, and mercury are the current metals of interest. Our intended audience would be environmental consultants and anyone involved in the use of fossil fuels. Questions that our dashboard will help users explore include:

- Which metals result in the highest mortality of freshwater plants?
- Which metal results in mortality with the lowest average concentration?
- Are some plant species more impacted by heavy metals than others?

While Alexa and Peyton have ecotoxicological interests, Marley is more interested in plant ecology. By developing a research question that combines these topics, we are able to engage different knowledge and skills from each member to produce an interdisciplinary dashboard. Additionally, since two members are from the python class, while one member is from the R class but has limited experience in python, we have chosen to use python for the project.

In terms of analytical quality, clarity, and professionalism, our audience can expect us to employ the data exploration skills that we have developed over the course of the semester. To maintain a clear, reproducible workflow, we will use ample comments to describe our code and keep a consistent format throughout the project (capitalization, spacing, indents, etc.).

2. Communication Plan and Collaborative Coding

Which channels will you use for routine communication, urgent questions, and sharing files or code? How quickly should teammates respond? When are members generally available to work, and when will you meet regularly? How will you document decisions and next steps so everyone stays informed?

- We will primarily communicate by text message both for routine communication and urgent questions, but will also address concerns in our weekly meetings
- File and code will be shared on github
- People should respond as quickly as they can (max time 24 hours)
- Generally everyone likes to work earlier in the day and does not like to do work after 9 pm
- We plan on meeting regularly on Tuesday afternoons from 2-3 and will schedule more meetings if needed
- We plan on doing most of the work together so we should all be informed about the decisions we made. If someone is not at a meeting then we will send them a text update.

3. Roles and Responsibilities

What interests each member about the project? What strengths do you bring, and what skills do you want to develop? How will you divide responsibilities for finding and preparing data, exploratory analysis, dashboard development, interpretation, documentation, and presentation? How will you ensure that everyone contributes meaningfully and understands the complete project? Identify who will review each other's work and how responsibilities may change as the project develops.

Peyton and Alexa have ecotoxicological interests (and are in the respective concentration). They are taking courses that provide context for the heavy metal pollutants that we are exploring. Marley has an interest in plant ecology and is in the Terrestrial and Freshwater concentration. We all want to develop our python skills for data exploration and science communication.

Responsibilities and expectations will be distributed among members during weekly team meetings (either during class on Thursdays or briefly on Tuesdays). While in-person, the intent is to work on weekly tasks together, with all members contributing to brainstorming codes and results. Additional individual tasks may follow in order to advance the project and complete tasks before deadlines. Everyone will be informed about the decisions made and why we are making them. Review of individual work will also happen during these scheduled meetings.

4. Operating Norms or Ground Rules

How will you prepare for and run meetings? How will you assign tasks, set internal deadlines, and track progress? What should teammates expect when someone completes work independently—for example, organized files, readable code, and documented data-cleaning decisions? What will you do if someone anticipates missing a meeting or deadline? How will you combine and review work before submitting it?

- We will text about what we are planning on focusing on during the meeting. This allows all members to have their individual contributions completed before meeting, which will allow us to adjust our plan for the upcoming week.
- We plan on doing most of the work together so will all set deadlines together and document decisions together
- If someone does work independently they should follow the same format as our starter code and update everyone by text when they are making changes and about what they did
- If someone is planning on missing a meeting the other two members will update them
- We are assigning a milestone lead to ensure that each task has been reviewed and refined before submitting.

5. Values and Behaviors

What does being a good teammate mean to your group? Agree on expectations for reliability, respectful communication, constructive feedback, and openness to different ideas. How will you create space for all members to contribute and ask for help? How will you address uneven participation or concerns about the quality of someone's work?

- Most of our work will be done in person together, so feedback and idea sharing can happen in person

        - We expect that everyone maintains respect and professionalism when providing criticism
        - We expect that new ideas are thoughtfully considered by all members before decisions are made
        - We expect that everyone contributes equally to the project
        - We expect that everyone is present for group meetings, or communicates if they have a conflict
- Since different components of the project play to different strengths within the group, we expect that some members will be more confident with certain tasks than others.
    - Knowing that each person has their strengths and weaknesses, we are able to openly communicate when we feel confident and when we need assistance with a certain task
    - Most of our communication will be in person, but we will have a group chat to voice concerns as well.
- Our weekly in person meetings will allow us to discuss individual contributions and address potential concerns.
    - Because we are a group of three, we feel that it will be easy to evenly distribute the workload, but are comfortable with having conversations if that is not the case.
    - If there are concerns over quality of work, we will reestablish the goals for the group to ensure that everyone is following consistent formatting that produces functional code

6. Resources and Constraints

What resources can support your work, including course materials, dataset documentation, instructors, TAs, and relevant subject-matter sources? What constraints should shape your scope—for example, the six-week timeline, missing data, differences in geographic or temporal coverage, technical skills, or other commitments?

Since follow-up with project partners may not be available, how will you resolve unanswered questions, document assumptions, and communicate limitations in the dashboard?

- We are planning on using a lot of the code from the first part of the (python) class in helping us do our project. Alexa will be provided with lab examples for python syntax.
- We tried to design a question that was realistic about our constraints and started with a question that could be expanded if we have more time
    - We started by looking at the mortality of fresh water plants, but could expand this to include salt water and terrestrial plants, or other species.
- We will talk to outside parties (instructors and TAs) as needed when we do not know how to answer a question
- In our dashboard we will try and be transparent about our limitations and have a section of the dashboard dedicated to this
    - Since our goal is to create reproducible, clear work, we are hopeful that this project could easily be expanded upon by individual members to answer any lingering questions.
    - Defining the limitations of the initial workflow could help guide the scope of project expansions.

7. Decision-Making Guidelines

How will you make decisions about research questions, datasets, analytical methods, and dashboard design? How will you ensure that relevant teammates are involved and that choices are supported by evidence? What will you do if you cannot reach consensus? When should you seek guidance from an instructor or TA?

- We will talk through decisions together and look at literature / ask the instructors on questions we are unsure of
- Since each member has variable experience with the research topic and coding medium, we will consider strengths and weaknesses of each person when assigning tasks.
- If we can't agree on a decision we will ask an instructor and get outside input
- An instructor could provide guidance if we reach a barrier that prevents us from effectively continuing the project. We will try to workshop these issues as a group first, then request help if we are stuck.

8. Goals and Objectives

What would a successful project look like? Identify a small set of concrete goals and how you will assess them. Consider whether the dashboard answers your research questions, uses appropriately prepared data, presents accurate and interpretable analyses, and is accessible and easy to navigate. What checks will you use to evaluate data quality, dashboard functionality, and the clarity of your findings? Distinguish essential deliverables from optional features.

Goals

- Visual outputs that effectively display the results of our research question
- A well formatted, reproducible, jupyter notebook that clearly demonstrates our workflow and results
- Easy to follow, well-labeled, clear code

To assess these goals, we will evaluate our outputs in group meetings, assign a milestone lead to do a final QA/QC, and have other groups review our outputs for quality, clarity and ease of use.

Essential Deliverables

- Clean dataset, functional dashboard
- Visually-appealing plots comparing mortality for each heavy metal
- We do not have plans at this time for other essential deliverables, but will continue to discuss this as we work with the data more.

Optional features

- Background information on heavy metals + fossil fuel combustion
- Identify species impacted by the most/least heavy metals

9. Schedule and Milestones

Milestone	Deadline for Group	Milestone Lead	Assignment

	Review	(Makes sure task is complete)	Deadline
Defining research questions and project scope.	Thurs Oct. 15	Group Effort	Tues Oct. 20
Assessing data feasibility.	Thurs Oct. 22	Alexa	Tues Oct. 27
Preparing and combining datasets.	Sunday Oct. 25	Peyton	Tues Oct. 27
Conducting exploratory analysis.	Sunday Nov. 1	Peyton	Tues Nov. 3
Building a dashboard prototype.	Sunday Nov. 8	Marley	Tues Nov. 10
Reviewing and testing the dashboard.	Sunday Nov. 15	Group Effort	Tues Nov. 17
Preparing the final presentation.	Tuesday Nov. 24	Group Effort	Thurs Dec. 3?